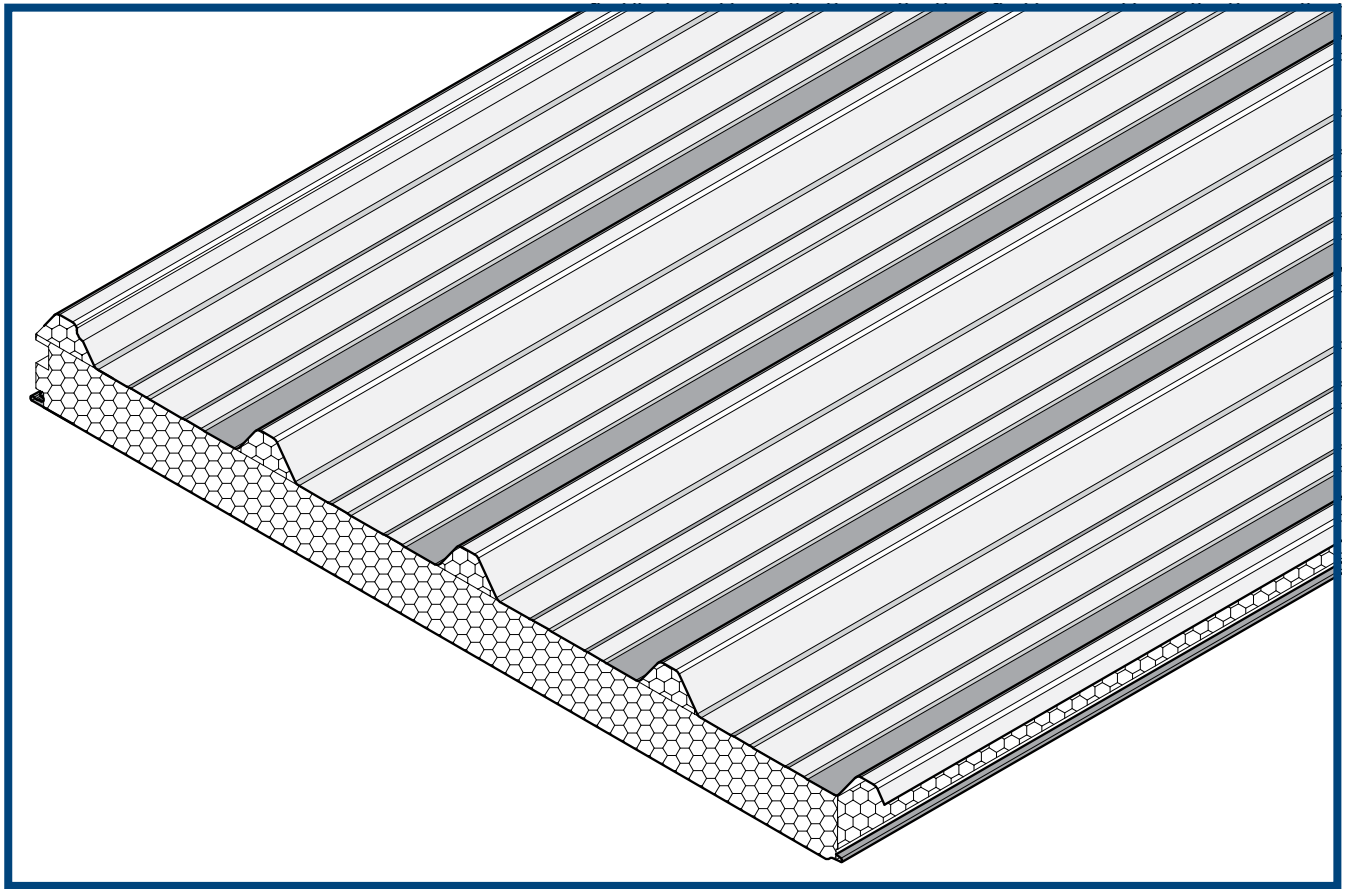




Solaris® roof panel is a long spanning, insulated, all-in-one roof product. Solaris® roof panel is energy efficient, hygienic, durable, aesthetically pleasing, fast and easy to install.

### Applications

#### Typical applications are:

- Industrial and Commercial roofing
- Home improvement patios
- Room enclosures
- Residential roofing

#### Commercial and industrial applications—Part J compliant energy efficient product:

- Air-conditioned offices
- Warehouses
- Factories
- Schools
- Shops
- Public buildings
- Food processing facilities and clean rooms.

### Features

- **Quick and easy** installation
- **Economical**
- **Low pitch** design with anti-capillary groove.
- **Long Spanning** panel, reducing structural support requirements.
- **Fire-retardant** treated core, BCA materials Group 2.
- **Void Filled Profile** for improved vermin proofing and insulation.

### Warranty

To the original purchaser, Retracom warrants for a period of fifteen (15) years, Solaris® insulated roofing panels (as supplied by Retracom) against de-lamination. Retracom shall provide free replacement (at our factory), product(s) deemed faulty by Retracom. This warranty expressly excludes parts or components not manufactured by Retracom\*. Such parts or components will be subject to their manufacturer's warranty.

\*Up to 15 years architectural panel roofing warranty - by application to BlueScope. The BlueScope warranties and application forms are downloadable on their website.



**firesmart**



**foodsmart**



**ecosmart**



**fibresmart**

EPSA Inc.  
Expanded Polystyrene  
Panel Manufacturers Group

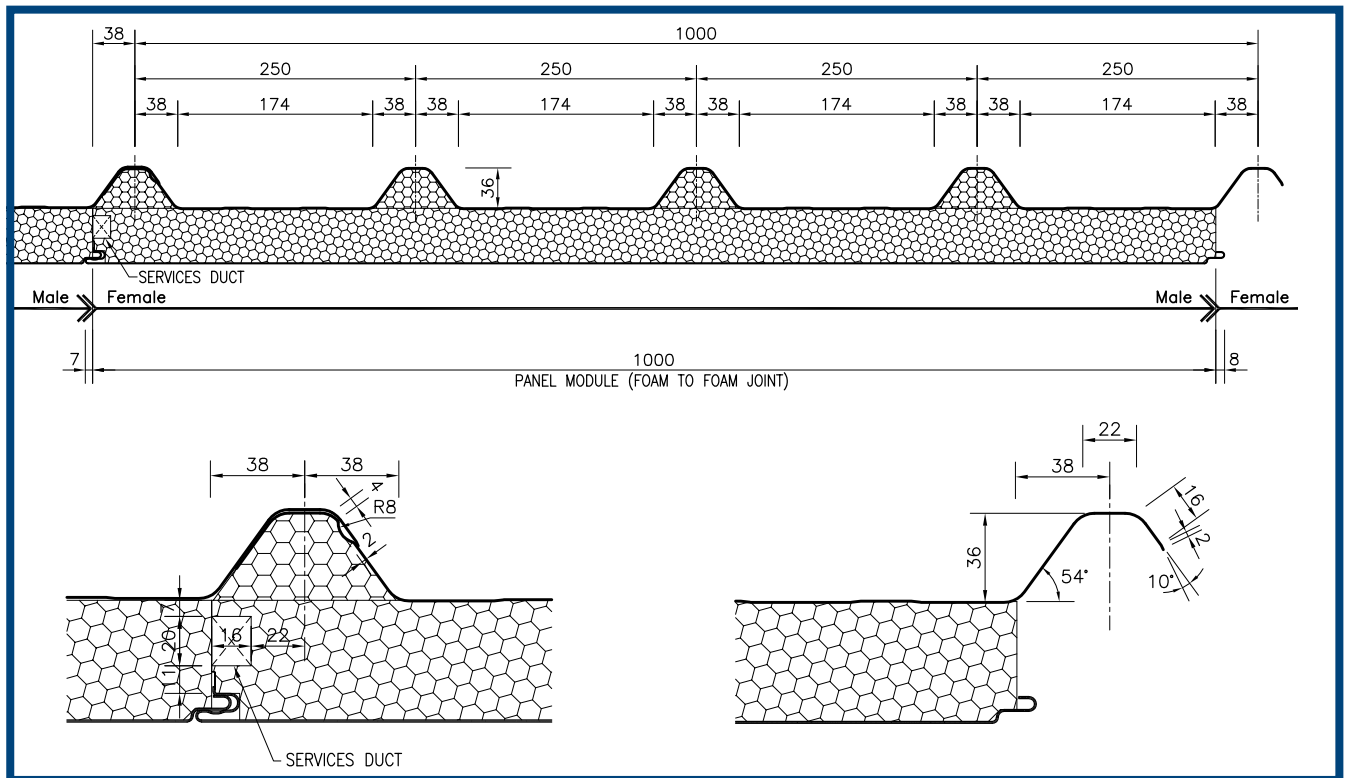


CERTIFIED  
PRODUCTS

Manufactured with

**Colorbond®** Steel

Last updated 08/05/11



### Specification

- **Width of module:** 1000mm
- **Thickness**  
*Standard:* 50mm, 75mm, 100mm, 125mm, 150mm
- **Length:**  
Minimum: 2400mm  
Maximum: 16,000mm
- **Rib profile:** Trapezoidal 36mm profile height.
- **Pitch (minimum):**  
Single span: 1.5°  
Multi span: 5°  
(See Roof Drainage Chart.)
- **Skins:**  
Roof: 0.42mm Hi-Tensile pre painted COLORBOND® steel.  
Ceiling: 0.5mm pre painted Colorbond® steel.
- **Stocked Colours:**  
*Roof side*  
Surfmist®  
Paper Bark®  
Classic Cream™  
Dune®  
Headland®  
Pale Eucalypt®  
*Ceiling side*  
Surfmist®
- **Flatness:** The core sheets are finger jointed and drum sanded to give the flattest surface possible.
- **Standard ceiling finish:** Strippable film to underside 'Smooth' with commercial flatness.
- **Core:** SL grade expanded polystyrene (EPS) with void filled profile.
- **Gutter cut back:** 60mm, 80mm, 100mm
- **BCA Materials Group Classification:** Group 2 (Group 1 is available with additional fixings.)
- **Spans:** Up to 9700mm (150mm Solaris® at 0.5kpa Ultimate design wind pressure.)
- **Jointing System:** Tongue and groove Retracom Flushline MkII joints.
- **Expansion joints:** Required as per roof installation code. See Solaris® drawing. Required every 16 metres max. Overhang 250mm at 5° minimum pitch.
- **Temperature:** Maximum skin temperature: 80°C.

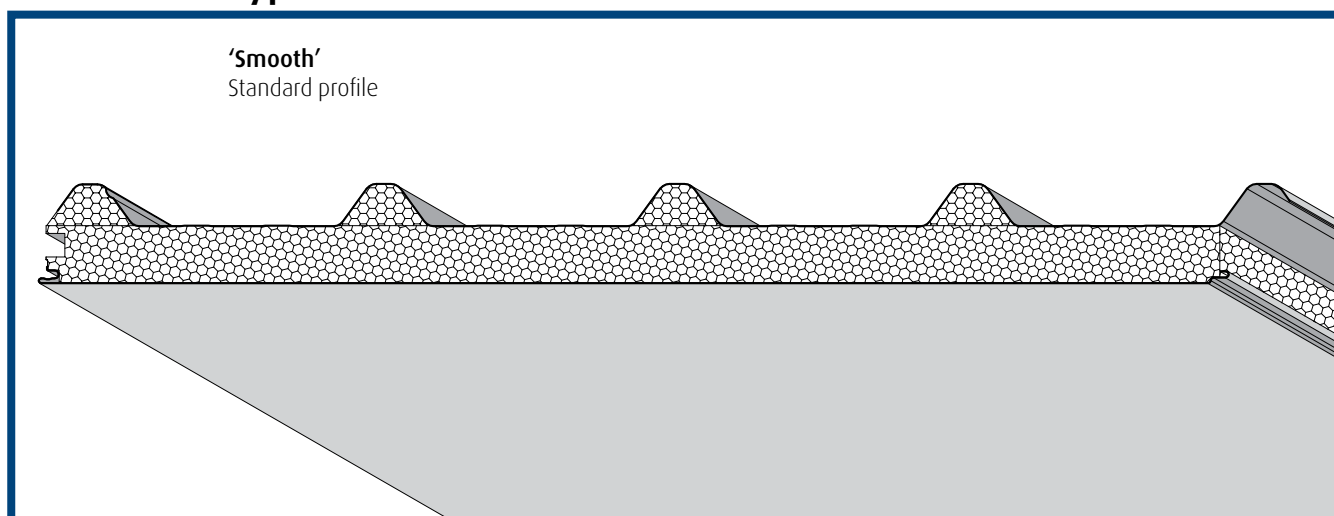
### Notes

\* Flatness appearance—'Commercial': Surface deformations are faintly detectable to the eye when observed in artificial or direct light and are apparent to the eye when observed in low incident light.

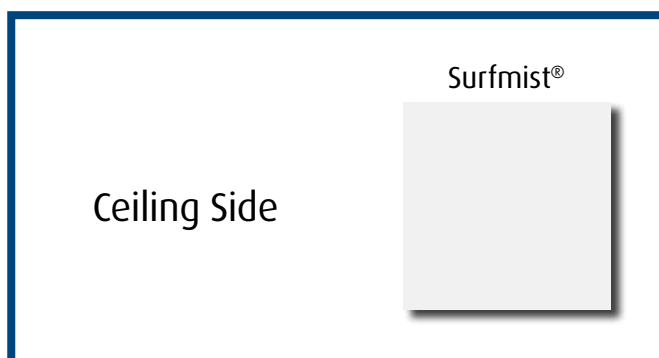
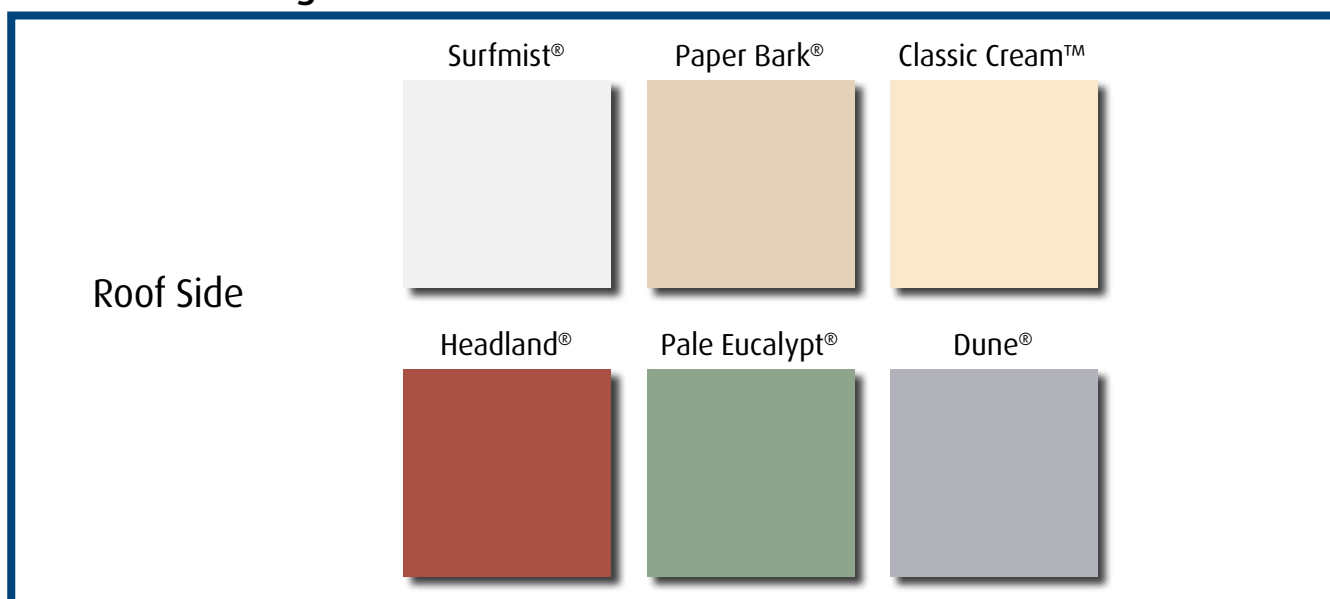
\*\* Subject to minimum quantities

## 2.2 Profile Type and Colours

### 2.2.1 Profile Type



### 2.2.2 Colour Range



#### Notes

- COLORBOND®, BlueScope and ® colour names are registered trade marks and ™ colour names are trade marks of BlueScope Steel Limited.
- The COLORBOND® steel colours shown have been reproduced to represent actual product colours as accurately as possible. However, we recommend checking your chosen colour against an actual sample of the product before purchasing, as light conditions and limitations of the printing process may affect colour tones. © 2008 BlueScope Steel Limited. ABN 16 000 011 058.

## 2.3 Wind Pressures for General Structures (kPa)

### Region A

|          | Terrain Category 2 |         |         | Terrain Category 2.5 |         |         | Terrain Category 3 |         |         |
|----------|--------------------|---------|---------|----------------------|---------|---------|--------------------|---------|---------|
|          | Wall               | Roof    |         | Wall                 | Roof    |         | Wall               | Roof    |         |
|          |                    | H/W<0.5 | H/W>1.0 |                      | H/W<0.5 | H/W>1.0 |                    | H/W<0.5 | H/W>1.0 |
| 5m high  | 1.01               | 1.21    | 1.51    | 0.92                 | 1.10    | 1.38    | 0.84               | 1.00    | 1.26    |
| 10m high | 1.22               | 1.46    | 1.82    | 1.02                 | 1.22    | 1.53    | 0.84               | 1.00    | 1.26    |
| 15m high | 1.34               | 1.61    | 2.01    | 1.14                 | 1.37    | 1.71    | 0.96               | 1.15    | 1.44    |

### Region B

|          | Terrain Category 2 |         |         | Terrain Category 2.5 |         |         | Terrain Category 3 |         |         |
|----------|--------------------|---------|---------|----------------------|---------|---------|--------------------|---------|---------|
|          | Wall               | Roof    |         | Wall                 | Roof    |         | Wall               | Roof    |         |
|          |                    | H/W<0.5 | H/W>1.0 |                      | H/W<0.5 | H/W>1.0 |                    | H/W<0.5 | H/W>1.0 |
| 5m high  | 1.61               | 1.94    | 2.42    | 1.48                 | 1.77    | 2.21    | 1.34               | 1.61    | 2.01    |
| 10m high | 1.95               | 2.34    | 2.92    | 1.63                 | 1.96    | 2.45    | 1.34               | 1.61    | 2.01    |
| 15m high | 2.15               | 2.58    | 3.22    | 1.83                 | 2.20    | 2.75    | 1.54               | 1.85    | 2.32    |

### Region C

|          | Terrain Category 2 |         |         | Terrain Category 2.5 |         |         | Terrain Category 3 |         |         |
|----------|--------------------|---------|---------|----------------------|---------|---------|--------------------|---------|---------|
|          | Wall               | Roof    |         | Wall                 | Roof    |         | Wall               | Roof    |         |
|          |                    | H/W<0.5 | H/W>1.0 |                      | H/W<0.5 | H/W>1.0 |                    | H/W<0.5 | H/W>1.0 |
| 5m high  | 3.51               | 4.16    | 5.20    | 2.98                 | 3.53    | 4.41    | 2.49               | 2.95    | 3.69    |
| 10m high | 3.89               | 4.61    | 5.76    | 3.47                 | 4.12    | 5.15    | 3.08               | 3.65    | 4.56    |
| 15m high | 4.45               | 5.28    | 6.60    | 3.97                 | 4.70    | 5.88    | 3.51               | 4.16    | 5.20    |

### Region D

|          | Terrain Category 2 |         |         | Terrain Category 2.5 |         |         | Terrain Category 3 |         |         |
|----------|--------------------|---------|---------|----------------------|---------|---------|--------------------|---------|---------|
|          | Wall               | Roof    |         | Wall                 | Roof    |         | Wall               | Roof    |         |
|          |                    | H/W<0.5 | H/W>1.0 |                      | H/W<0.5 | H/W>1.0 |                    | H/W<0.5 | H/W>1.0 |
| 5m high  | 5.66               | 6.71    | 8.39    | 4.80                 | 5.69    | 7.11    | 4.01               | 4.76    | 5.95    |
| 10m high | 6.27               | 7.43    | 9.29    | 5.60                 | 6.64    | 8.30    | 4.97               | 5.89    | 7.36    |
| 15m high | 7.18               | 8.51    | 10.64   | 6.40                 | 7.58    | 9.48    | 5.66               | 6.71    | 8.39    |

## 2.4 Wind Load Conversions (non-cyclonic)

| Wind Classification (Domestic) | Region & Category (Commercial/Industry) |
|--------------------------------|-----------------------------------------|
| N1 (W28)                       | Reg A, Cat 3                            |
| N2 (W33)                       | Reg A, Cat 2.5 - Reg B, Cat 3           |
| N3 (W41)                       | Reg A, Cat 2, Reg B, Cat 2.5            |
| N4 (W50)                       | Reg B, Cat 2                            |

#### Notes

- Pressures are for interlocking sandwich panels used for external cladding of a base structure.
- H/W is the reaction of the average roof height to the minimum width of the building.
- For values of h/d between 0.5 and 1.0, use linear interpolation.
- Full in.
- Assumptions:
  - Md = 1.0 - Sandwich Panel is not an critical element of the base structure.
  - Ms = Mt = 1.0 - Relatively flat and unshielded site.
  - Kl = 1.0 - Panel is interlocked to distribute pressures, and is a structural element.
  - Ka = Kc = 1.0
  - External Pressures - Walls: Cpe = +0.7, -0.5, Roof: Cpe = +0.2, -0.9 (h/d<0.5), -1.5(h/d>1.0)
  - Internal Pressure - Region A & B - Cpi = +0.2, -0.3, Region C & D: Cpi = +0.7, -0.65

## 2.5 Allowable Panel Spans (mm)

| Pressure (kPa) | Panel Thickness |      |       |       |       |
|----------------|-----------------|------|-------|-------|-------|
|                | 50mm            | 75mm | 100mm | 125mm | 150mm |
| 0.5            | 5600            | 7700 | 8500  | 9250  | 9700  |
| 0.75           | 5600            | 6650 | 7300  | 7550  | 7900  |
| 1              | 5250            | 5750 | 6300  | 6550  | 6850  |
| 1.25           | 4700            | 5150 | 5650  | 5850  | 6100  |
| 1.5            | 4300            | 4700 | 5150  | 5350  | 5600  |
| 1.75           | 4000            | 4350 | 4750  | 4950  | 5150  |
| 2              | 3650            | 4050 | 4450  | 4600  | 4850  |
| 2.5            | 2900            | 3650 | 4000  | 4150  | 4300  |
| 3              | 2450            | 3300 | 3650  | 3750  | 3950  |
| 3.5            | 2100            | 3050 | 3350  | 3500  | 3650  |
| 4              | 1800            | 2750 | 3150  | 3250  | 3400  |
| 4.5            | 1600            | 2450 | 2950  | 3050  | 3200  |
| 5              | 1450            | 2200 | 2800  | 2900  | 3050  |
| 6              | 1200            | 1800 | 2450  | 2650  | 2800  |
| 7              | 1050            | 1550 | 2100  | 2450  | 2550  |
| 8              | 900             | 1350 | 1800  | 2300  | 2400  |

## 2.6 Design Pressure (kPa)

| Panel Span (mm) | Panel Thickness |      |       |       |       |
|-----------------|-----------------|------|-------|-------|-------|
|                 | 50mm            | 75mm | 100mm | 125mm | 150mm |
| 2000            | 3.6             | 5.5  | -     | -     | -     |
| 2500            | 2.9             | 4.4  | 5.8   | 6.5   | -     |
| 3000            | 2.4             | 3.6  | 4.4   | 4.8   | 5.2   |
| 3500            | 2.1             | 2.7  | 3.2   | 3.5   | 3.8   |
| 4000            | 1.7             | 2.1  | 2.5   | 2.7   | 2.9   |
| 4500            | 1.3             | 1.6  | 1.9   | 2.1   | 2.3   |
| 5000            | 1.1             | 1.3  | 1.6   | 1.7   | 1.8   |
| 5500            | 0.9             | 1.1  | 1.3   | 1.4   | 1.5   |
| 6000            | -               | 0.9  | 1.1   | 1.2   | 1.3   |
| 6500            | -               | 0.7  | 0.9   | 1     | 1.1   |
| 7000            | -               | 0.6  | 0.8   | 0.8   | 0.9   |
| 7500            | -               | 0.5  | 0.7   | 0.7   | 0.8   |
| 8000            | -               | -    | 0.6   | 0.6   | 0.7   |
| 8500            | -               | -    | 0.5   | 0.5   | 0.6   |
| 9000            | -               | -    | -     | -     | 0.5   |

### Notes

- Table 1 shows the maximum allowable spacing of roof supports, and table 2 shows maximum design pressures, for single span or continuous span Solaris® roofing panel.
- Pressures in table 1 and 2 are for ultimate design wind pressure with 1:500 year return period, applied in either direction. Permanent, long-term, snow, and other loadings are not allowed for. Pressure do not include the self weight of the panel.
- Maximum deflections of span/150 for a 1:20 year return period are used for wind loading, in regions A, B, C or D, as per AS1170.1.
- Largest spans are checked for deflections of span/200, and structural capacity, for self weight and 0.25kPa/1.1kN live load as per AS1170.1.
- Largest spans may develop thermal bowing of up to span/180, for a 35°C temperature gradient (-10°C to 60°C external, 25°C internal). Thermal effects are not taken into account for panel spans.
- Expansion joints are to be included as required, but at no more than 16m spacing.
- Panels are to be fixed to supports with 14kg tek screws with washers to each rib. The pullout capacity of the screws shall be checked.
- Table 3 shows the allowable bearing capacity of the Solaris Panel, for design pressure acting towards the support, per metre of support length.

## 2.7 Bearing Capacity (kN/m)

| Thickness (mm) | Support Location | Bearing Width |      |      |       |       |       |       |
|----------------|------------------|---------------|------|------|-------|-------|-------|-------|
|                |                  | 50mm          | 70mm | 90mm | 100mm | 150mm | 200mm | 250mm |
| 50             | internal         | 4.2           | 5.3  | 6.4  | 7.0   | 9.8   | 12.6  | 15.4  |
|                | external         | 3.5           | 4.6  | 5.7  | 6.3   | 9.1   | 11.9  | 14.7  |
| 75             | internal         | 4.9           | 6.0  | 7.1  | 7.7   | 10.5  | 13.3  | 16.1  |
|                | external         | 3.9           | 5.0  | 6.1  | 6.7   | 9.5   | 12.3  | 15.1  |
| 100            | internal         | 5.6           | 6.7  | 7.8  | 8.4   | 11.2  | 14.0  | 16.8  |
|                | external         | 4.2           | 5.3  | 6.4  | 7.0   | 9.8   | 12.6  | 15.4  |
| 125            | internal         | 6.3           | 7.42 | 8.54 | 9.1   | 11.9  | 14.7  | 17.5  |
|                | external         | 4.55          | 5.67 | 6.79 | 7.35  | 10.15 | 12.95 | 15.75 |
| 150            | internal         | 7             | 8.12 | 9.24 | 9.8   | 12.6  | 15.4  | 18.2  |
|                | external         | 4.9           | 6.02 | 7.14 | 7.7   | 10.5  | 13.3  | 16.1  |

### Notes

- Table 2.5 shows maximum allowable spacing of roof supports, and table 2 shows maximum design pressures, for single span or continuous span Solaris® roofing panel.
- Pressures in table 2.5 and 2.6 are for ultimate design wind pressure with 1:500 year return period, applied in either direction. Permanent, long-term, snow, and other loadings are not allowed for. Pressure does not include the self weight of the panel
- Maximum deflections of span/150 for a 1:20 year return period are used for wind loading, in regions A, B, C or D, as per AS1170.2
- Largest spans are checked for deflections of span/200, and structural capacity, for self weight and 0.25kPa/1.1kN live load as per AS1170.1
- Largest spans may develop thermal bowing of up to span/180, for a 35°C temperature gradient (-10°C to 60°C external, 25°C internal). Thermal effects are not taken into account for panel spans.
- Expansion joints are to be included as required, but at no more than 16m spacing.
- Panels are to be fixed to supports with 14kg tek screws with washers to each rib. The pullout capacity of the screws shall be checked.
- Table 2.7 shows the allowable bearing capacity of the Solaris® Panel, for design pressure acting towards the support, per metre of support length.

## 2.8 Spans and overhangs (mm) Domestic awnings only

- Select the Solaris® panel thickness from the span table below for the wind condition on site (refer building inspector if required).
- Refer to drawings PR-09-008-01 to 09 for complete configuration and engineering of the patio roof system.

| Wind Rating | Solaris® Thickness | 3 sides open A |          | 3 sides open B |          | 2 sides open |          | 1 side open |          | Enclosed |          |
|-------------|--------------------|----------------|----------|----------------|----------|--------------|----------|-------------|----------|----------|----------|
|             |                    | Span           | Overhang | Span           | Overhang | Span         | Overhang | Span        | Overhang | Span     | Overhang |
| <b>N1</b>   | 50mm               | 7500           | 900      | 6900           | 900      | 5800         | 900      | 5300        | 900      | 5500     | 900      |
|             | 75mm               | 8200           | 900      | 7600           | 900      | 6400         | 900      | 5800        | 900      | 6100     | 900      |
|             | 100mm              | 8900           | 900      | 8300           | 900      | 6900         | 900      | 6300        | 900      | 6600     | 900      |
|             | 125mm              | 9000           | 900      | 8600           | 900      | 7200         | 900      | 6600        | 900      | 6900     | 900      |
|             | 150mm              | 9400           | 900      | 9000           | 900      | 7500         | 900      | 6900        | 900      | 7200     | 900      |
| <b>N2</b>   | 50mm               | 6400           | 900      | 5900           | 900      | 4900         | 900      | 4500        | 900      | 4700     | 900      |
|             | 75mm               | 7000           | 900      | 6500           | 900      | 5400         | 900      | 4900        | 900      | 5100     | 900      |
|             | 100mm              | 7600           | 900      | 7000           | 900      | 5900         | 900      | 5400        | 900      | 5600     | 900      |
|             | 125mm              | 7900           | 900      | 7300           | 900      | 6100         | 900      | 5600        | 900      | 5800     | 900      |
|             | 150mm              | 8300           | 900      | 7700           | 900      | 6400         | 900      | 5800        | 900      | 6100     | 900      |
| <b>N3</b>   | 50mm               | 5100           | 900      | 4700           | 900      | 3900         | 900      | 3600        | 900      | 3800     | 900      |
|             | 75mm               | 5600           | 900      | 5200           | 900      | 4300         | 900      | 3900        | 900      | 4100     | 900      |
|             | 100mm              | 6100           | 900      | 5600           | 900      | 4700         | 900      | 4300        | 900      | 4500     | 900      |
|             | 125mm              | 6300           | 900      | 5900           | 900      | 4900         | 900      | 4500        | 900      | 4700     | 900      |
|             | 150mm              | 6600           | 900      | 6100           | 900      | 5100         | 900      | 4700        | 900      | 4900     | 900      |
| <b>N4</b>   | 50mm               | 4200           | 900      | 3900           | 900      | 3200         | 800      | 3000        | 800      | 3100     | 800      |
|             | 75mm               | 4600           | 900      | 4200           | 900      | 3500         | 900      | 3200        | 900      | 3400     | 900      |
|             | 100mm              | 5000           | 900      | 4600           | 900      | 3900         | 900      | 3500        | 900      | 3700     | 900      |
|             | 125mm              | 5200           | 900      | 4800           | 900      | 4000         | 900      | 3700        | 900      | 3800     | 900      |
|             | 150mm              | 5400           | 900      | 5000           | 900      | 4200         | 900      | 3800        | 900      | 4000     | 900      |
| <b>N5</b>   | 50mm               | 3400           | 850      | 3200           | 900      | 2700         | 700      | 2300        | 600      | 2500     | 600      |
|             | 75mm               | 3800           | 900      | 3500           | 900      | 2900         | 700      | 2700        | 700      | 2800     | 700      |
|             | 100mm              | 4100           | 900      | 3800           | 900      | 3200         | 800      | 2900        | 700      | 3000     | 700      |
|             | 125mm              | 4300           | 900      | 4000           | 900      | 3300         | 800      | 3000        | 750      | 3200     | 800      |
|             | 150mm              | 4500           | 900      | 4100           | 900      | 3500         | 850      | 3200        | 800      | 3300     | 800      |
| <b>C1</b>   | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |
|             | 75mm               | 5600           | 900      | 5200           | 900      | 4300         | 900      | 3900        | 900      | 3400     | 850      |
|             | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |
| <b>C2</b>   | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |
|             | 75mm               | 4600           | 900      | 4200           | 900      | 3500         | 850      | 3200        | 800      | 2800     | 700      |
|             | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |
| <b>C3</b>   | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |
|             | 75mm               | 3800           | 900      | 3500           | 875      | 2900         | 700      | 2700        | 650      | -        | -        |
|             | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |
| <b>C4</b>   | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |
|             | 75mm               | 3200           | 800      | 3000           | 750      | 2200         | 550      | -           | -        | -        | -        |
|             | 75mm               | -              | -        | -              | -        | -            | -        | -           | -        | -        | -        |

### Notes to "3 Sides Open"

- Case "A" can be used for:
  - Roofs attached to or under the eaves level of a single-story residence elevated no more than 1.0m above ground.
  - Roofs at ground level attached to a double-story residences.
- Case "B" can be used for:
  - Elevated roofs for a single story residences.
  - Roofs attached to the eaves of a double-story residence.
  - Where the average height of the roofs is no more than 85% of the average height of the roof of the residence it is attached to.
- For situations that are outside of Case "B", use the span table for "Two Sides Open".

## 2.9 Maximum roof lengths

|           | 1.5°**<br>roof slope | 2.0°**<br>roof slope | 3.0°**<br>roof slope | 5.0°<br>roof slope | 7.5°<br>roof slope | 10.0°<br>roof slope |
|-----------|----------------------|----------------------|----------------------|--------------------|--------------------|---------------------|
| 150mm/hr* | 16m                  | 16m                  | 16m                  | 210m               | 250m               | 290m                |
| 200mm/hr* | 16m                  | 16m                  | 16m                  | 160m               | 190m               | 220m                |
| 250mm/hr* | 15m                  | 16m                  | 16m                  | 120m               | 160m               | 170m                |
| 300mm/hr* | 14m                  | 16m                  | 16m                  | 100m               | 120m               | 140m                |
| 400mm/hr* | 12m                  | 16m                  | 16m                  | 80m                | 95m                | 110m                |
| 450mm/hr* | 10m                  | 16m                  | 16m                  | 60m                | 70m                | 80m                 |

\* Peak rainfall intensity

\*\* Continous sheet required from ridge to gutter

## 2.10 Stacking Information

| Panel Thickness               | Panel Quantity and Height (mm), Nominated by Length |        |    |        |     |        |     |        | Weight m <sup>2</sup> |
|-------------------------------|-----------------------------------------------------|--------|----|--------|-----|--------|-----|--------|-----------------------|
|                               | 4m                                                  | Height | 6m | Height | 10m | Height | 12m | Height |                       |
| 50mm                          | 17                                                  | 1215   | 10 | 710    | 6   | 440    | 5   | 405    | 10.0 kg               |
| 75mm                          | 12                                                  | 1145   | 9  | 900    | 5   | 530    | 5   | 530    | 10.3 kg               |
| 100mm                         | 10                                                  | 1210   | 8  | 975    | 5   | 655    | 5   | 655    | 10.7 kg               |
| 125mm                         | 8                                                   | 1145   | 8  | 1145   | 5   | 740    | 5   | 740    | 11.0kg                |
| (Average Max. Weight = 600kg) |                                                     |        |    |        |     |        |     |        |                       |

## 2.11 Panel Properties

| Panel Thickness (mm)                                |     |      |      |     |      |
|-----------------------------------------------------|-----|------|------|-----|------|
|                                                     | 50  | 75   | 100  | 125 | 150  |
| Weight (Kg/m <sup>2</sup> )                         | 10  | 10.3 | 10.7 | 11  | 11.3 |
| R Value (m <sup>2</sup> K/W) Built up Systems: Roof | 1.7 | 2.4  | 3.0  | 3.7 | 4.3  |